

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.(BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 3 Duration : 1 Hour Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

*I SHAILAJA S(192511026)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

Constraints:

- i. D1 cannot work Night shift
- ii. D2 must work before D3 (Morning < Afternoon < Night)
- iii. Only one doctor per shift
- iv. D3 cannot work Morning
- v. All doctors must be assigned exactly one shift

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

Step 1: Define Variables and Domains

Doctors: D1, D2, D3

Shifts: Morning (M), Afternoon (A), Night (N)

Initial Domain:

- D1 → {M, A} (cannot take Night)
- D2 → {M, A, N}
- D3 → {A, N} (cannot take Morning)

Step 2: Apply Backtracking Search

Start with D1:

- Try D1 = Morning

Now assign D2:

- Possible values: {Afternoon, Night} (Morning already assigned)
- Try D2 = Afternoon

Now assign D3:

- Remaining shift = Night
- Check constraints:
 - D3 ≠ Morning ✓

- D2 before D3 → Afternoon < Night ✓
- Valid assignment found

Check all constraints:

- D1 ≠ Night ✓
- D2 before D3 ✓
- One doctor per shift ✓
- D3 ≠ Morning ✓
- All doctors assigned ✓

Step 3: Backtracking Tree (Simplified)

1. D1 = Morning
→ D2 = Afternoon
→ D3 = Night ✓ (Valid Solution)

(No need to backtrack further since solution found)

Final Valid Schedule:

Doctor	Assigned Shift
D1	Morning
D2	Afternoon
D3	Night

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Cost of each move = 1
- iii. Cannot pass through obstacles
- iv. Use Manhattan Distance heuristic

Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost.

Step 1: Grid Representation

Example Grid:

S 0 X 0 0

0 0 X 0 0

0 0 0 0 0

X X 0 X 0

0 0 0 G 0

- S = Start (0,0)
- G = Goal (4,3)
- X = Obstacle

Step 2: BFS Concept

- BFS explores nodes level by level
- Uses a Queue (FIFO)
- Guarantees optimal path when cost is equal

Step 3: Node Expansion (Step-by-Step)

Initial State:

Queue = [(0,0)]

Visited = {(0,0)}

Step 1: Expand (0,0)

Possible moves → (0,1), (1,0)

Queue = [(0,1), (1,0)]

Step 2: Expand (0,1)

Moves → (1,1)

Queue = [(1,0), (1,1)]

Step 3: Expand (1,0)

Moves → (2,0)

Queue = [(1,1), (2,0)]

Step 4: Expand (1,1)

Moves → (2,1)

Queue = [(2,0), (2,1)]

Step 5: Expand (2,0)

Moves → (2,1) (already visited)

Queue = [(2,1)]

Step 6: Expand (2,1)

Moves → (2,2)

Queue = [(2,2)]

Step 7: Expand (2,2)

Moves → (2,3), (3,2)

Queue = [(2,3), (3,2)]

Step 8: Expand (2,3)
Moves $\rightarrow$ (2,4)
Queue = [(3,2), (2,4)]

Step 9: Expand (3,2)
Moves $\rightarrow$ (4,2)
Queue = [(2,4), (4,2)]

Step 10: Expand (4,2)
Moves $\rightarrow$ (4,3) $\rightarrow$ GOAL FOUND
Queue = [(2,4), (4,3)]

Step 4: Path Tracing

Backtracking from Goal:

$(4,3) \leftarrow (4,2) \leftarrow (3,2) \leftarrow (2,2) \leftarrow (2,1) \leftarrow (1,1) \leftarrow (0,1) \leftarrow (0,0)$

Step 5: Optimal Path

$S \rightarrow (0,1) \rightarrow (1,1) \rightarrow (2,1) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (4,2) \rightarrow G$

Step 6: Cost Calculation

- Total steps = 7
- Cost per step = 1

Total Cost = 7

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

- a) Analyze all the given constraints and explain how they affect the robot's decision-making process.
- b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
- c) Demonstrate how the robot applies AI concepts such as:
 - Intelligent agents
 - Rationality
 - Environment type
- d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

a) Analysis of Constraints and Their Impact on Decision-Making

The robot moves only in four directions, limiting actions to nearby cells. Obstacles block paths, so it must avoid invalid moves. Since the environment is partially observable, it can only sense adjacent cells, creating uncertainty. Movement cost is 1, while risky zones cost 3, so the robot must balance safety and efficiency. As it updates knowledge dynamically, decisions must be made in real time instead of planning the full path in advance.

b) Solution Approach Using Uniform Cost Search with Online Search Behavior

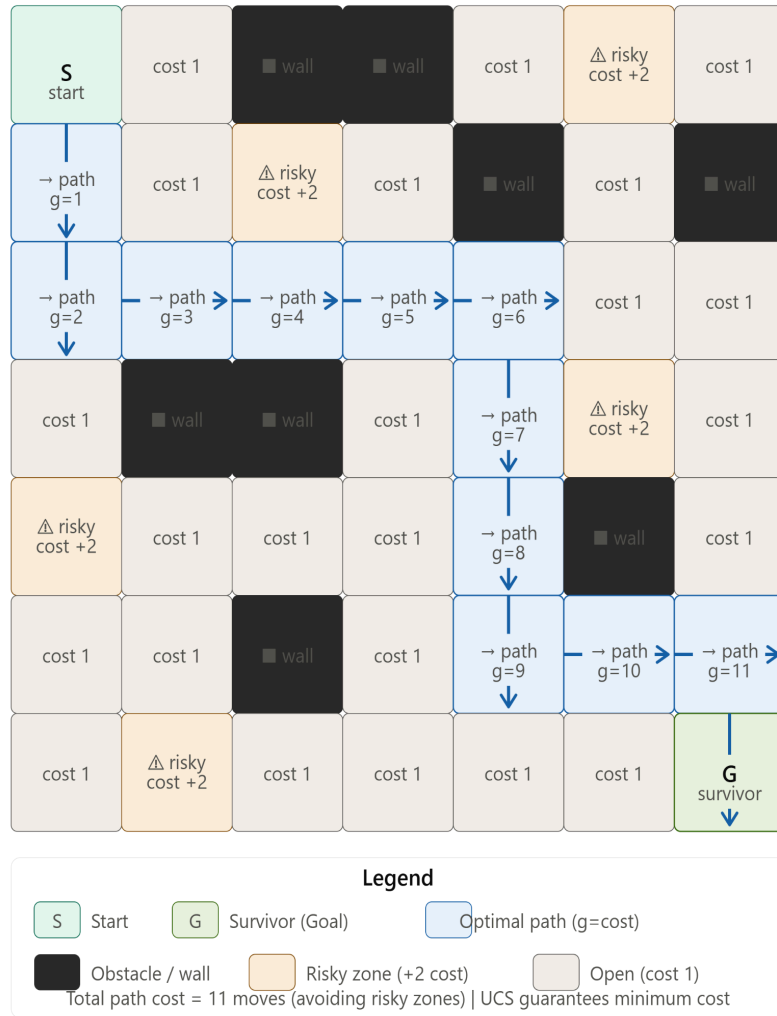
The robot uses Uniform Cost Search with online updates. It starts at S, senses nearby cells, and adds valid moves to a priority queue based on cost. It always selects the lowest-cost path, updates knowledge after each move, and adjusts decisions if better paths are found. This process continues until the goal G is reached, ensuring optimal and adaptive navigation.

c) Application of AI Concepts

The robot acts as an intelligent agent by sensing and acting to reach its goal. It is rational because it chooses actions that minimize cost. The environment is partially observable, dynamic, discrete, and uncertain. The robot follows a goal-based and utility-based approach to achieve efficient and safe movement.

d) Justification of the Chosen Approach

Uniform Cost Search ensures the least-cost path, which is important due to different movement costs. Since the environment is not fully known, combining it with online search helps the robot adapt to new information. This approach handles obstacles, uncertainty, and dynamic updates effectively, making it suitable for rescue operations.



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly

Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided